EQUIVALENCES OF COHERENT MODULE CATEGORIES ON NOETHERIAN SCHEMES COME FROM ISOMORPHISMS AND INVERTIBLE MODULES

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0FZL: sources/stacks-chow/tags.txt, source lines 19956–19956. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 0GPJ: sources/stacks-chow/tags.txt, source lines 20829–20829. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X and Y be Noetherian schemes. Let $F : Coh(\mathcal{O}_X) \to Coh(\mathcal{O}_Y)$ be an equivalence of categories. Then there is an isomorphism $f : Y \to X$ and an invertible $\mathcal{O}_Y$-module $\mathcal{L}$ such that $F(\mathcal{F}) = f^*\mathcal{F} \otimes \mathcal{L}$.*

Proof. By Lemma 0FZL we obtain a unique functor $F' : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Y)$ extending F. The same is true for the quasi-inverse of F and by the uniqueness we conclude that F' is an equivalence. By Proposition 0GPJ we find an isomorphism $f : Y \to X$ and an invertible $\mathcal{O}_Y$-module $\mathcal{L}$ such that $F'(\mathcal{F}) = f^*\mathcal{F} \otimes \mathcal{L}$. Then f and $\mathcal{L}$ work for F as well. □

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